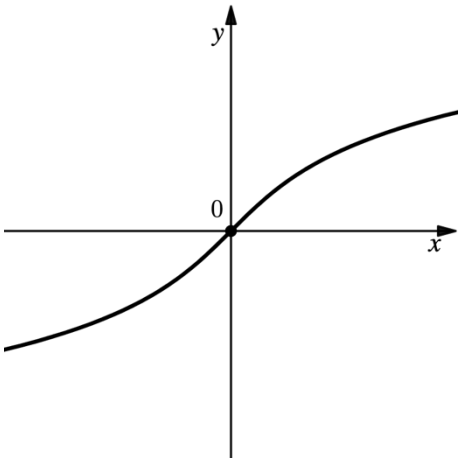
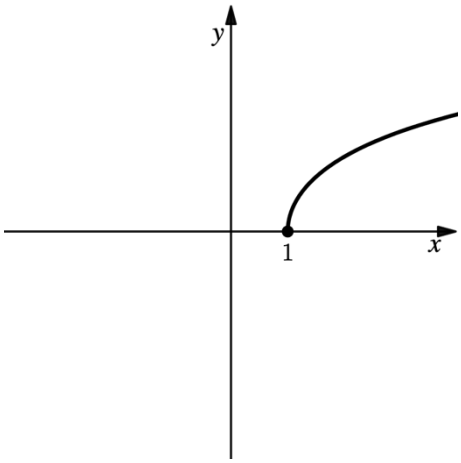
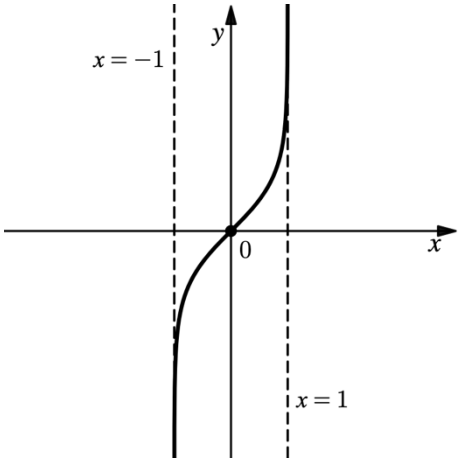
**1) a)**



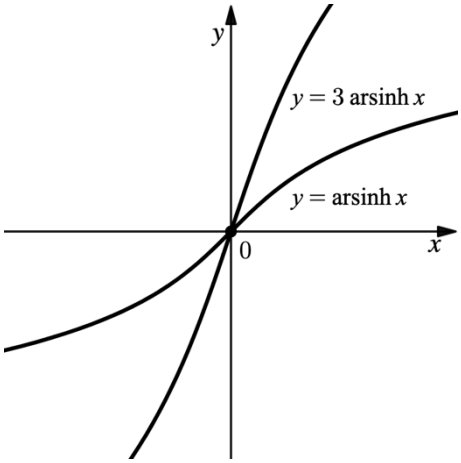

**1) b)**



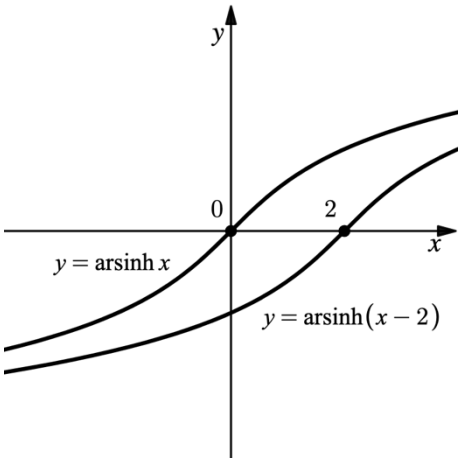

**1) c)**



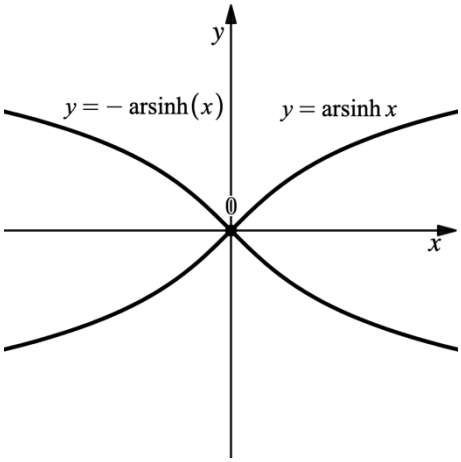
**2) a)**


y
y = 3 arsinh x
y = arsinh x
0
x

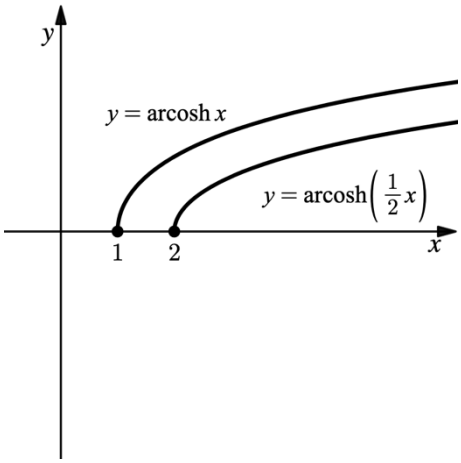
**2) b)**


y
0
2
x
y = arsinh x
y = arsinh(x − 2)

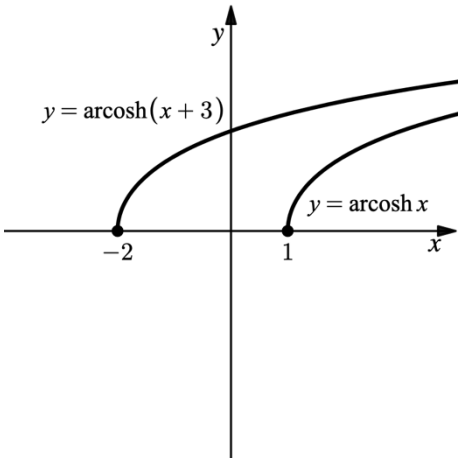
**2) c)**


y
y = − arsinh(x)
y = arsinh x
0
x

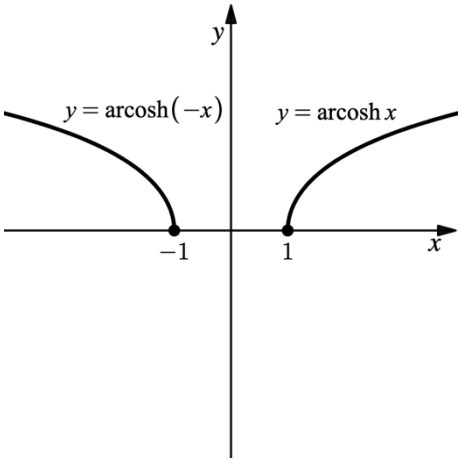
**2) d)**


y
y = arcosh x
y = arcosh(1/2 x)
1
2
x

**2) e)**


y
y = arcosh(x + 3)
y = arcosh x
−2
1
x

**2) f)**


y
y = arcosh(−x)
y = arcosh x
−1
1
x

**3) a)**

$y = \operatorname{arsinh} x + b$

$\left(-\frac{21}{20}, 0\right)$:

$0 = \operatorname{arsinh}\left(-\frac{21}{20}\right) + b$

$\Rightarrow 0 = \ln\left(-\frac{21}{20} + \sqrt{\left(-\frac{21}{20}\right)^2 + 1}\right) + b$

$\Rightarrow 0 = \ln\left(-\frac{21}{20} + \frac{29}{20}\right) + b$

$\Rightarrow b = -\ln\frac{2}{5} = \ln\frac{5}{2}$

$\therefore y = \operatorname{arsinh} x + \ln\frac{5}{2}$

**3) b)**

$y = \operatorname{arsinh}(x - a)$

$\left(0, \ln\left(\frac{\sqrt{13}-3}{2}\right)\right)$:

$\ln\left(\frac{\sqrt{13}-3}{2}\right) = \operatorname{arsinh}(-a)$

$\Rightarrow \ln\left(\frac{\sqrt{13}-3}{2}\right) = \ln\left(-a + \sqrt{(-a)^2 + 1}\right)$

$\Rightarrow \ln\left(\frac{\sqrt{13}-3}{2}\right) = \ln\left(-a + \sqrt{a^2 + 1}\right)$

$\Rightarrow \frac{\sqrt{13}-3}{2} = -a + \sqrt{a^2 + 1}$

$\Rightarrow a + \frac{\sqrt{13}-3}{2} = \sqrt{a^2 + 1}$

$\Rightarrow \left(a + \frac{\sqrt{13}-3}{2}\right)^2 = a^2 + 1$

$\Rightarrow a^2 + \left(\sqrt{13} - 3\right)a + \frac{11-3\sqrt{13}}{2} = a^2 + 1$

$\Rightarrow \left(\sqrt{13} - 3\right)a + \frac{11-3\sqrt{13}}{2} = 1$

$\Rightarrow \left(\sqrt{13} - 3\right)a = \frac{3\sqrt{13}-9}{2}$

$\Rightarrow a = \frac{3\sqrt{13}-9}{2\sqrt{13}-6} = \frac{3}{2}$

$\therefore y = \operatorname{arsinh}\left(x - \frac{3}{2}\right)$

**4) a)**

$y = \operatorname{arcosh} x + b$

$\left(\frac{13}{5}, 0\right)$:

$0 = \operatorname{arcosh}\frac{13}{5} + b$

$\Rightarrow 0 = \ln\left(\frac{13}{5} + \sqrt{\left(\frac{13}{5}\right)^2 - 1}\right) + b$

$\Rightarrow 0 = \ln\left(\frac{13}{5} + \frac{12}{5}\right) + b$

$\Rightarrow 0 = \ln 5 + b$

$\Rightarrow b = -\ln 5$

$\therefore y = \operatorname{arcosh} x - \ln 5$

**4) b)**

$y = \operatorname{arcosh}(x - a)$

$\left(0, \ln\left(3 + 2\sqrt{2}\right)\right)$:

$\ln\left(3 + 2\sqrt{2}\right) = \operatorname{arcosh}(-a)$

$\Rightarrow \ln\left(3 + 2\sqrt{2}\right) = \ln\left(-a + \sqrt{(-a)^2 - 1}\right)$

$\Rightarrow \ln\left(3 + 2\sqrt{2}\right) = \ln\left(-a + \sqrt{a^2 - 1}\right)$

$\Rightarrow 3 + 2\sqrt{2} = -a + \sqrt{a^2 - 1}$

$\Rightarrow a + 3 + 2\sqrt{2} = \sqrt{a^2 - 1}$

$\Rightarrow \left(a + 3 + 2\sqrt{2}\right)^2 = a^2 - 1$

$\Rightarrow a^2 + \left(6 + 4\sqrt{2}\right)a + 12\sqrt{2} + 17 = a^2 - 1$

$\Rightarrow \left(6 + 4\sqrt{2}\right)a + 12\sqrt{2} + 17 = -1$

$\Rightarrow \left(6 + 4\sqrt{2}\right)a = -18 - 12\sqrt{2}$

$\Rightarrow a = -\frac{18 + 12\sqrt{2}}{6 + 4\sqrt{2}} = -3$

$\therefore y = \operatorname{arcosh}(x + 3)$

**5) a)**

$y = \operatorname{artanh} x$ has horizontal asymptotes at $x = -1$ and $x = 1$

$y = \operatorname{artanh} x \mapsto y = \operatorname{artanh}(x + 5) - 3$ is a translation by the vector $\begin{pmatrix} -5 \\ -3 \end{pmatrix}$, so the vertical asymptotes are translated 5 units to the left.

$\therefore y = \tanh(x + 5) - 3$ has vertical asymptotes at $x = -6$ and $x = -4$

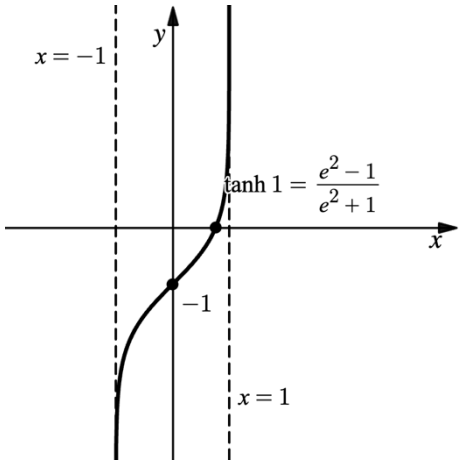
**5) b)**

$y = \operatorname{artanh} x$ has horizontal asymptotes at $x = -1$ and $x = 1$

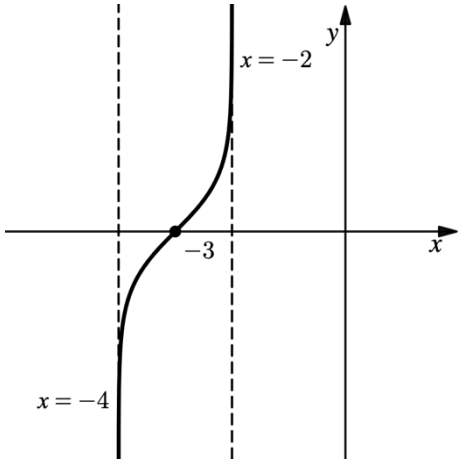
$y = \operatorname{artanh} x \mapsto y = \operatorname{artanh}\left(x - \frac{1}{3}\right) + 2$ is a translation by the vector $\begin{pmatrix} \frac{1}{3} \\ 2 \end{pmatrix}$, so the vertical asymptotes are translated $\frac{1}{3}$ units to the right.

$\therefore y = \tanh\left(x - \frac{1}{3}\right) + 2$ has vertical asymptotes at $x = -\frac{2}{3}$ and $x = \frac{4}{3}$

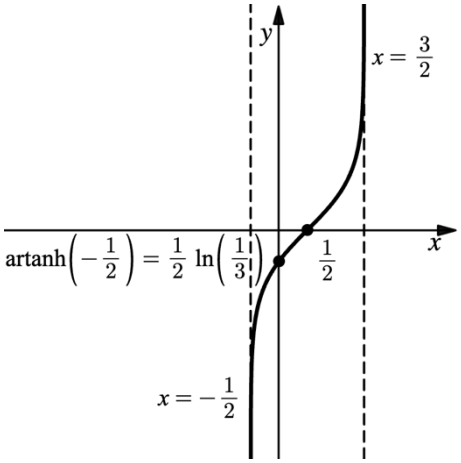
**6) a)**



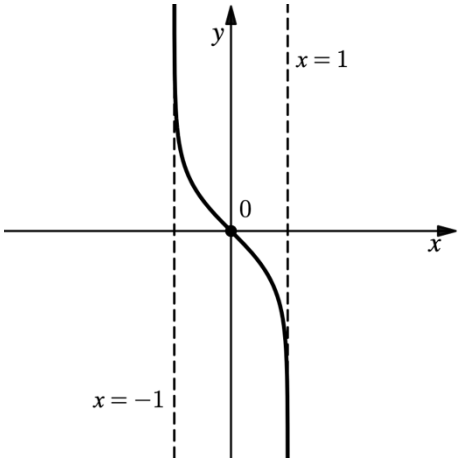

**6) b)**



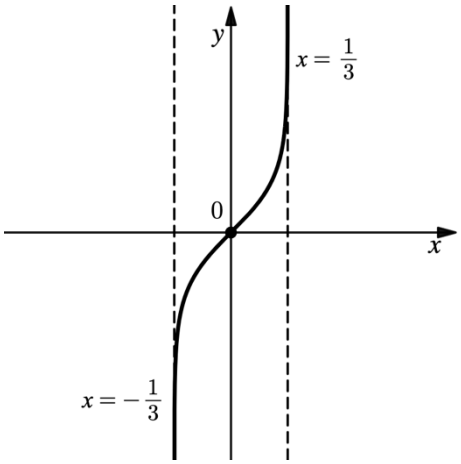
**6) c)**


y
$x = \frac{3}{2}$
$\text{artanh}\left(-\frac{1}{2}\right) = \frac{1}{2}\ln\left(\frac{1}{3}\right)$
$\frac{1}{2}$
x
$x = -\frac{1}{2}$

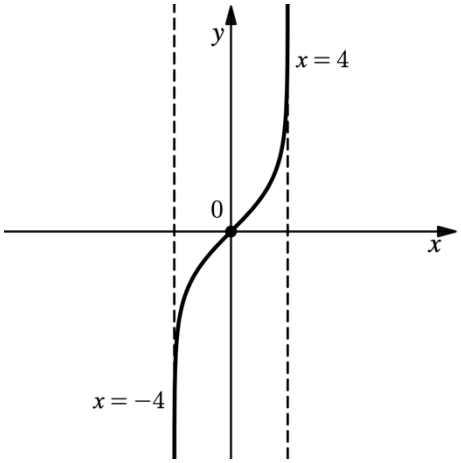
**6) d)**


y
$x = 1$
0
x
$x = -1$

**7) a)**


y
$x = \frac{1}{3}$
0
x
$x = -\frac{1}{3}$

**7) b)**




**8) a)**

$\frac{-\frac{13}{8}+\frac{3}{8}}{2} = -\frac{5}{8}$

$y = \operatorname{artanh}\left(x + \frac{5}{8}\right)$

**8) b)**

$y = \operatorname{artanh} x + b$

$\left(-\frac{4}{5}, 0\right)$:

$0 = \operatorname{artanh}\left(-\frac{4}{5}\right) + b$

$\Rightarrow 0 = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{4}{5}\right)}{1-\left(-\frac{4}{5}\right)}\right) + b$

$\Rightarrow 0 = \frac{1}{2}\ln\left(\frac{1+\left(-\frac{4}{5}\right)}{1-\left(-\frac{4}{5}\right)}\right) + b$

$\Rightarrow 0 = \frac{1}{2}\ln\frac{1}{9} + b$

$\Rightarrow 0 = \ln\frac{1}{3} + b$

$\Rightarrow b = -\ln\frac{1}{3} = \ln 3$

$\therefore y = \operatorname{artanh} x + \ln 3$

**8) c)**

$y = \text{artanh}(x - a)$

$\left(0, \ln\left(\frac{\sqrt{2}}{2}\right)\right)$:

$\ln\left(\frac{\sqrt{2}}{2}\right) = \text{artanh}(-a)$

$\Rightarrow \ln\left(\frac{\sqrt{2}}{2}\right) = \frac{1}{2}\ln\left(\frac{1+(-a)}{1-(-a)}\right)$

$\Rightarrow 2\ln\left(\frac{\sqrt{2}}{2}\right) = \ln\left(\frac{1-a}{1+a}\right)$

$\Rightarrow \ln\left(\frac{1}{2}\right) = \ln\left(\frac{1-a}{1+a}\right)$

$\Rightarrow \frac{1}{2} = \frac{1-a}{1+a}$

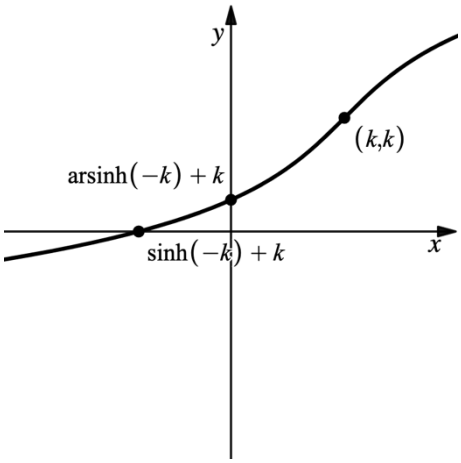
$\Rightarrow 1 + a = 2 - 2a$

$\Rightarrow 3a = 1$

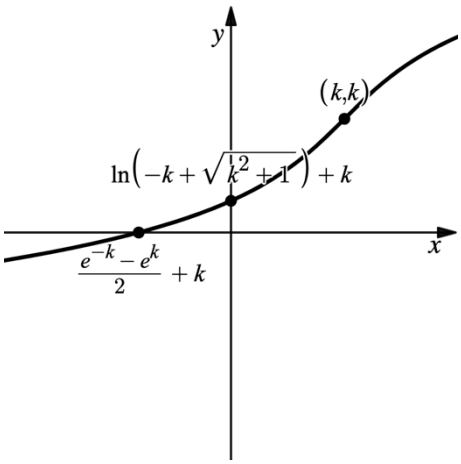
$\Rightarrow a = \frac{1}{3}$

$\therefore y = \text{artanh}\left(x - \frac{1}{3}\right)$

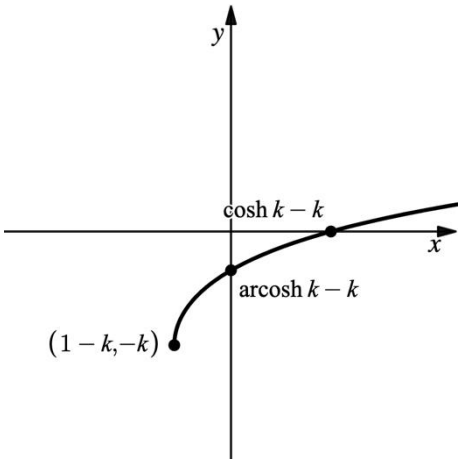
**9) a)**



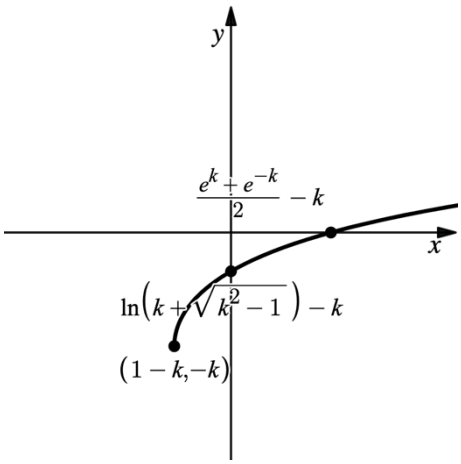

Or



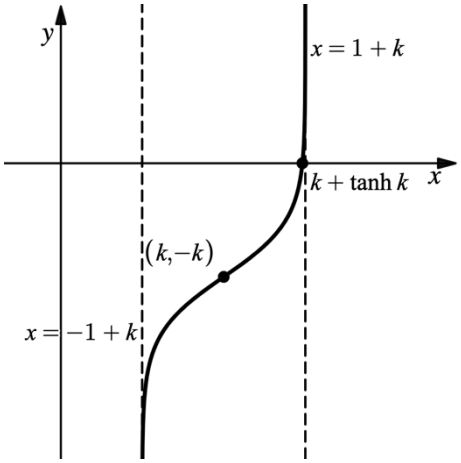
**9) b)**


y
cosh k − k
x
arcosh k − k
(1 − k,−k)

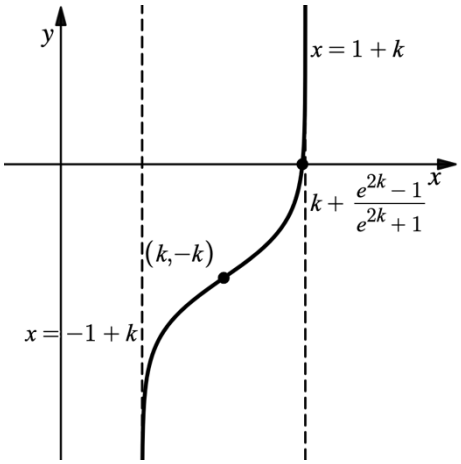
Or


y
$\frac{e^k + e^{-k}}{2} - k$
x
$\ln\left(k + \sqrt{k^2 - 1}\right) - k$
(1 − k,−k)

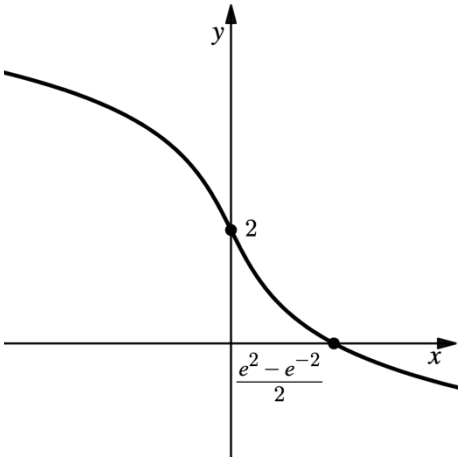
**9) c)**



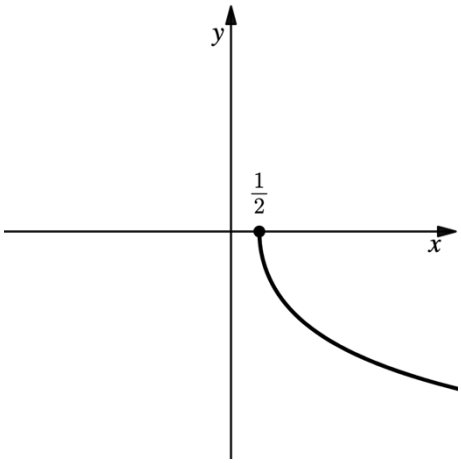

Or



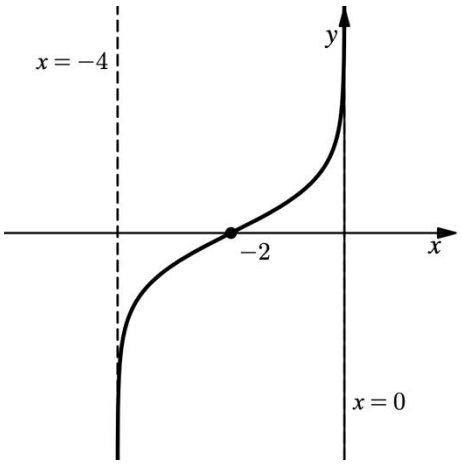

**10) a)**



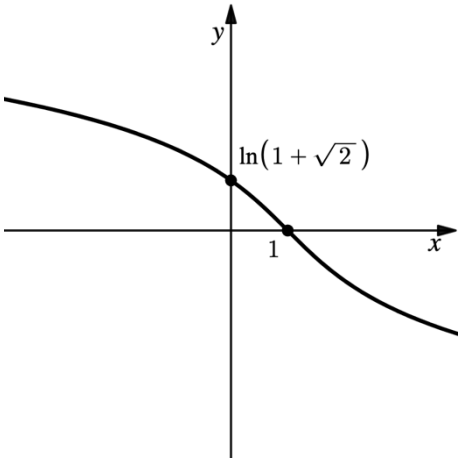
**10) b)**



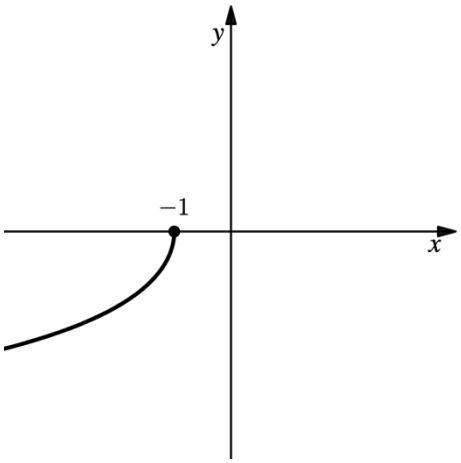

**10) c)**



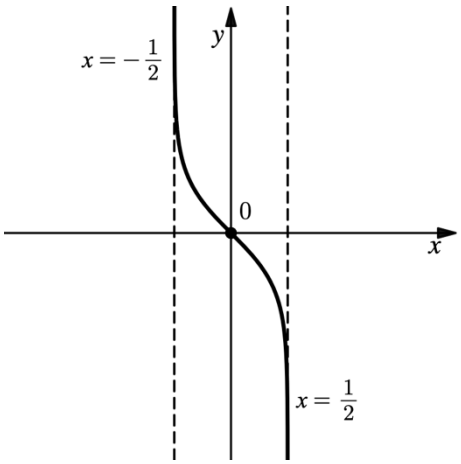

**10) d)**



**10) e)**

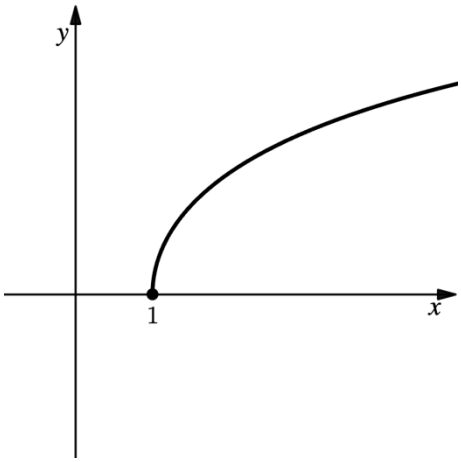



**10) f)**

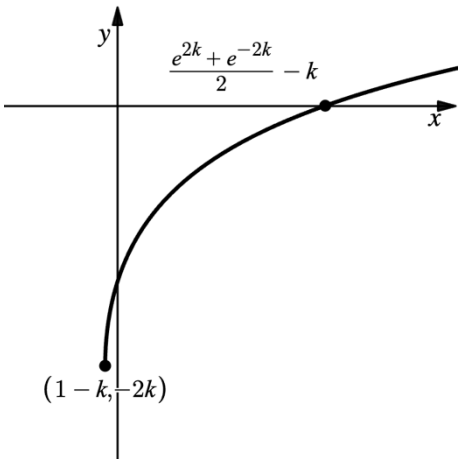


**11) a)**

First consider $y = \operatorname{arcosh} x$:

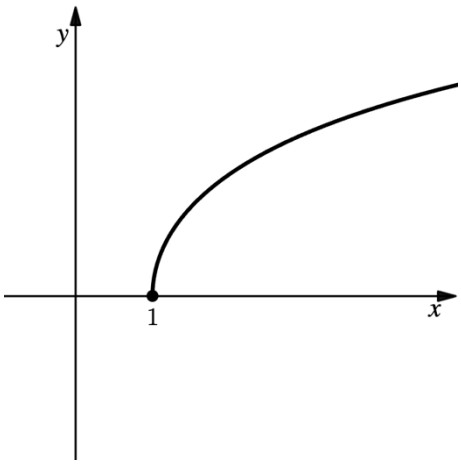



Translate by $\begin{pmatrix} -k \\ -2k \end{pmatrix}$ to get $y = \operatorname{arcosh}(x + k) - 2k$:

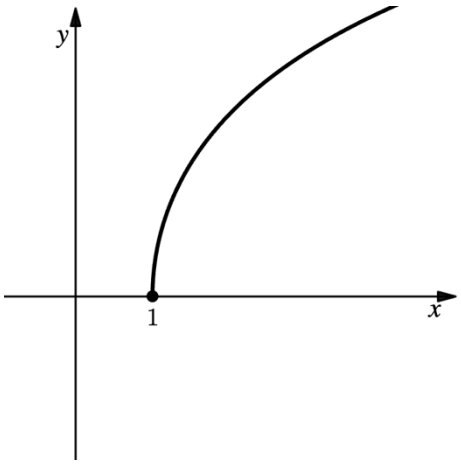



But this isn't quite right as the $x$-intercept should be $\frac{\mathrm{e}^2+\mathrm{e}^{-2}}{2} - k$ instead of $\frac{\mathrm{e}^{2k}+\mathrm{e}^{-2k}}{2} - k$

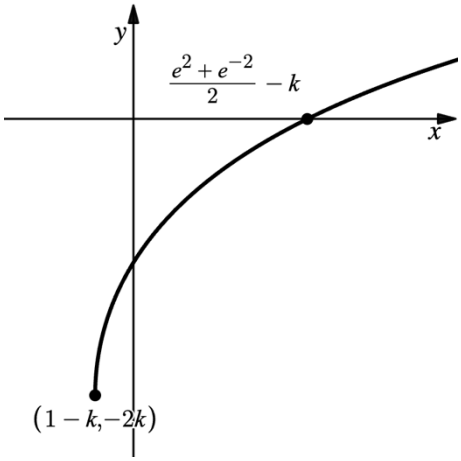
So start again with $y = \operatorname{arcosh} x$:




Then Stretch, parallel to the $y$-axis, by scale factor $k$ to get $y = k \operatorname{arcosh} x$:



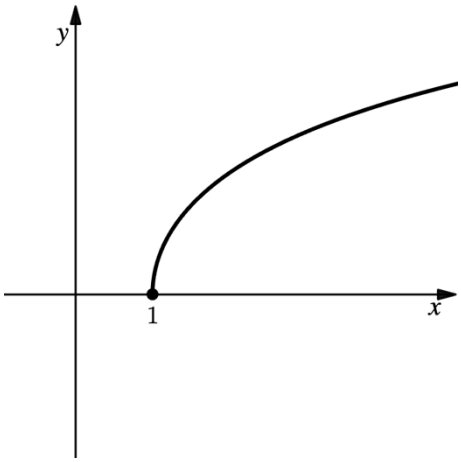

Translate by $\begin{pmatrix} -k \\ -2k \end{pmatrix}$ to get $y = k\,\text{arcosh}(x + k) - 2k$:



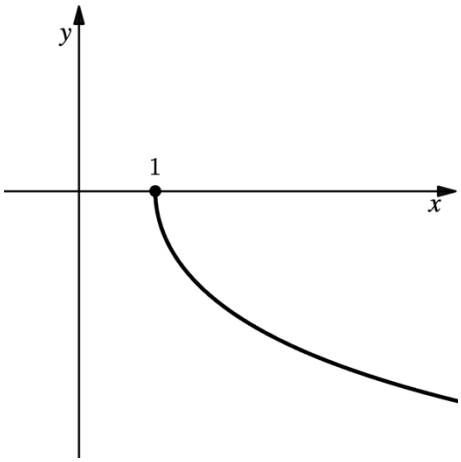

$\therefore y = k\,\text{arcosh}(x + k) - 2k$

**11) b)**

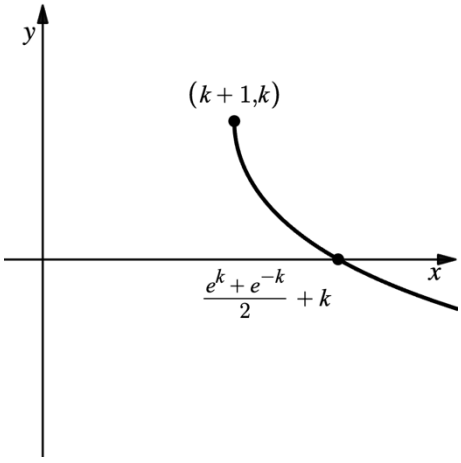
First consider $y = \operatorname{arcosh} x$:




Reflect in the $x$-axis to get $y = -\operatorname{arcosh} x$:

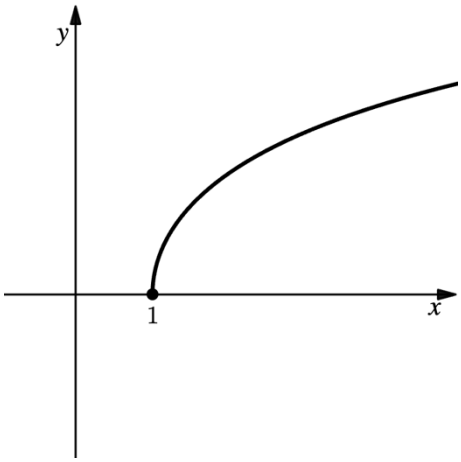



Translate by $\begin{pmatrix} k \\ k \end{pmatrix}$ to get $y = -\operatorname{arcosh}(x-k) + k$:

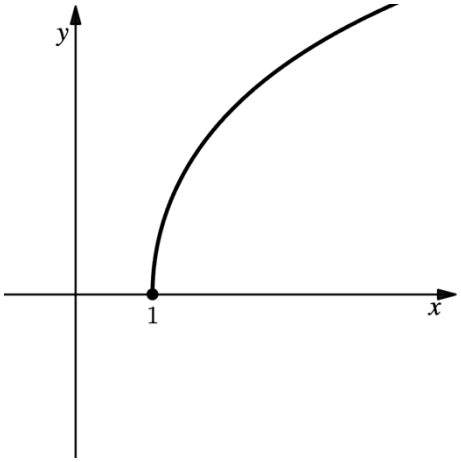



But this isn't quite right as the $x$-intercept should be $\frac{\mathrm{e}+\mathrm{e}^{-1}}{2} - k$ instead of $\frac{\mathrm{e}^{k}+\mathrm{e}^{-k}}{2} - k$

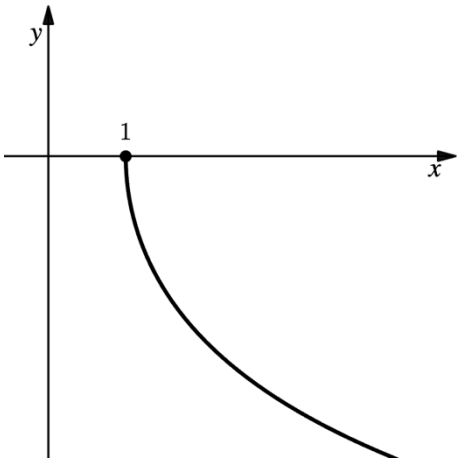
So start again with $y = \operatorname{arcosh} x$:



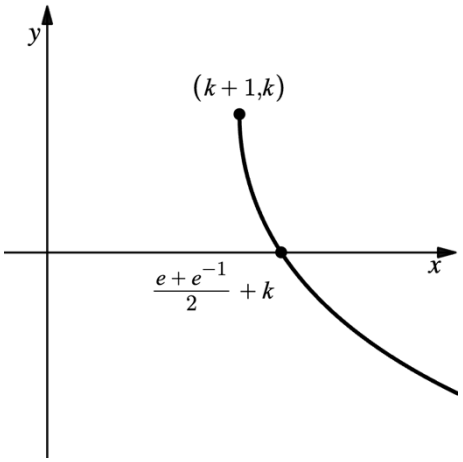

Then Stretch, parallel to the $y$-axis, by scale factor $k$ to get $y = k\,\text{arcosh}\,x$:




Reflect in the $x$-axis to get $y = -k\,\text{arcosh}\,x$:

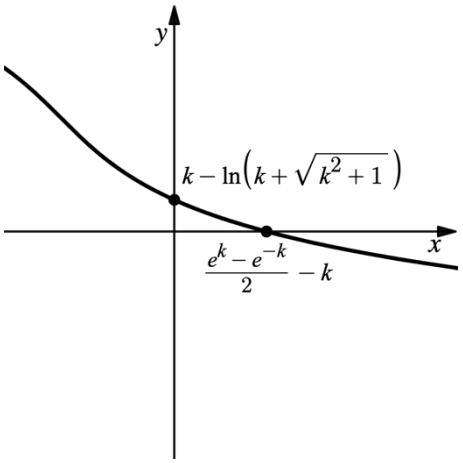



Translate by $\begin{pmatrix} k \\ k \end{pmatrix}$ to get $y = -k\,\text{arcosh}(x-k) + k$:

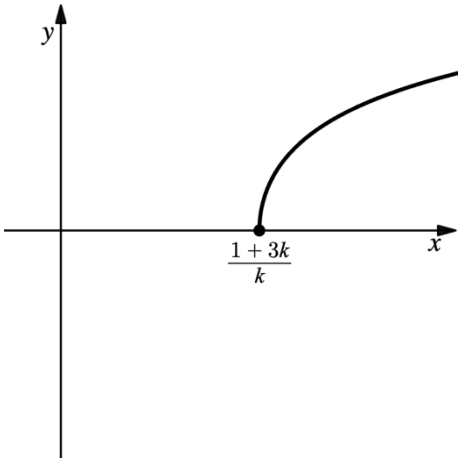



$\therefore y = -k\,\text{arcosh}(x-k)+k$

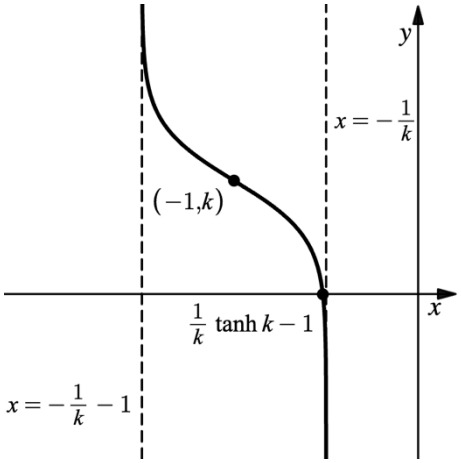
**12) a)**



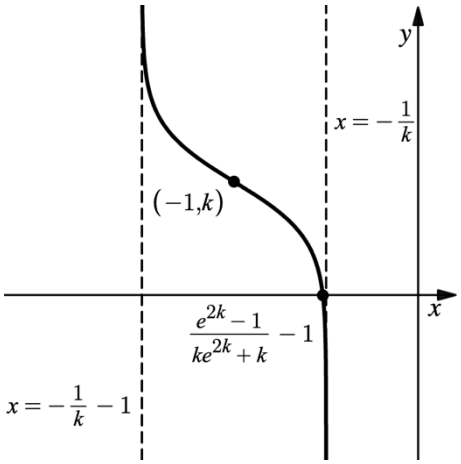

**12) b)**



**12) c)**


y
$x = -\frac{1}{k}$
$(-1,k)$
$\frac{1}{k}\tanh k - 1$
x
$x = -\frac{1}{k} - 1$

Or


y
$x = -\frac{1}{k}$
$(-1,k)$
$\frac{e^{2k}-1}{ke^{2k}+k} - 1$
x
$x = -\frac{1}{k} - 1$